Handout 4

Chapter 19

Elasticity

The law of demand illustrates that price and quantity demanded are inversely related, ceteris paribus. But it doesn't tell us by what percentage the quantity demanded changes as price changes. Elasticity of demand does. **Price elasticity of demand**, which is a measure of the responsiveness of quantity demanded to changes in price. More specifically, it addresses the percentage change in quantity demanded for a given percentage change in price.

Derivation of formula for elasticity demand

$$E_d = \frac{\text{Percentage change in quantity demand}}{\text{Percentage change in price}} = \frac{\%\Delta Q_d}{\%\Delta P}$$

$$= \frac{\frac{\Delta Q_d}{Q_d} x100}{\frac{\Delta P}{P} x100} = \frac{\frac{\Delta Q_d}{Q_d}}{\frac{\Delta P}{P}} = \frac{\Delta Q_d}{Q} . \frac{P}{\Delta P} = \frac{\Delta Q_d}{\Delta P} . \frac{P}{Q_d}$$

P = Price of the commodity
Q_d = quantity demanded

$E_d = \frac{\Delta Q_d}{\Delta P} \frac{P}{Q_d}$, Formula of elasticity of demand.

$E_d = \frac{\Delta Q_d}{\Delta P} \frac{P}{Q_d}$, (P and Q_d are average price and average quantity demanded to avoid misleading results)

Determine Elasticity of demand from the following information.

P	Q_d
2	10
1	25

Determine E_d.

$$E_d = \frac{\Delta Q_d}{\Delta P}\frac{P}{Q_d}$$

$$\Delta Q = Q_2 - Q_1 = 25 - 10 = 15$$
$$\Delta P = P_2 - P_1 = 1 - 2 = -1$$
$$P = \frac{1+2}{2} = 1.5$$
$$Q_d = \frac{10+25}{2} = 17.5$$

$$E_d = \frac{\Delta Q_d}{\Delta P}\frac{P}{Q_d}$$

$$= \frac{15}{-1} x \frac{1.5}{17.5} = -1.2857$$ (Estimated elasticity of demand)

The negative sign is due to demand law. We take absolute value.

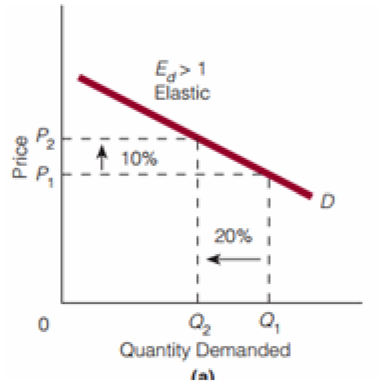
i.e., $E_d = 1.2857$

TYPES OF ELASTICITY OF DEMAND

1. Elastic demand, $E_d > 1$: Means, % change in quantity demanded > % change in price.
2. Inelastic demand, $E_d < 1$: Means, % change in quantity demanded < % change in price.
3. Unitary elastic demand, $E_d = 1$: Means, % change in quantity demanded = % change in price.
4. Perfectly elastic demand, $E_d = \infty$: Means, a little change in price brings huge change in quantity demanded.
5. Perfectly inelastic demand, $E_d = 0$: Means, a change in price brings no change in quantity demanded.

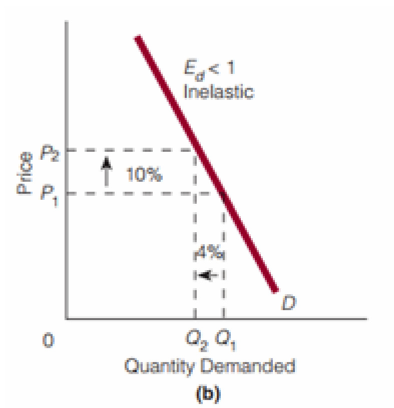
Figures for different types of elasticity of demand

1. Elastic demand



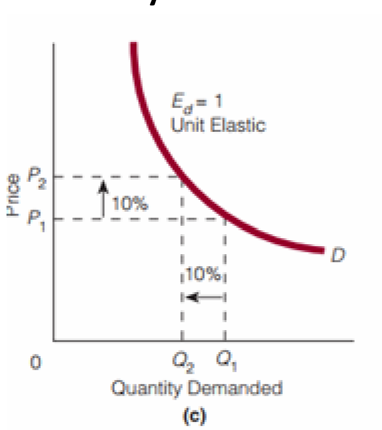
Interpretation: % change in quantity demanded > % change in price.

2. Inelastic demand



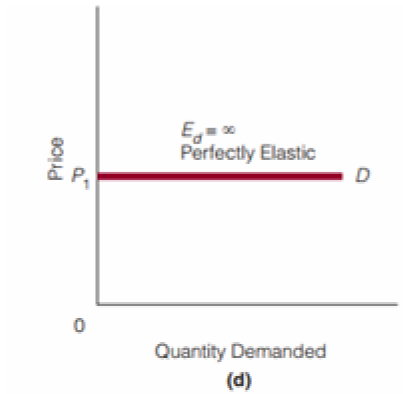
Interpretation: % change in quantity demanded < % change in price.

3. Unitary elastic demand



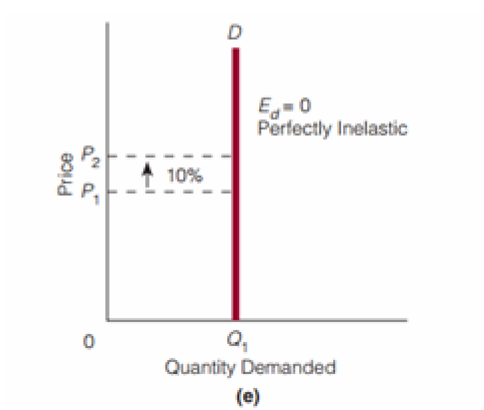
Interpretation: % change in quantity demanded = % change in price.

4. Perfectly elastic demand



Interpretation: Means, a little change in price brings huge change in quantity demanded.

5. Perfectly inelastic demand



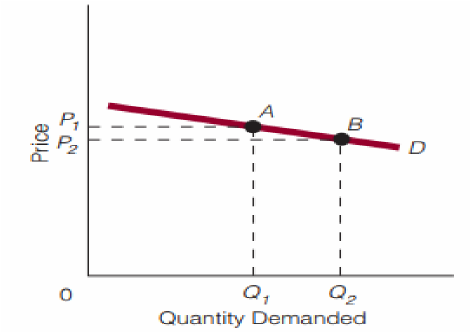
Interpretation: a change in price brings no change in quantity demanded.

Price Elasticity of Demand and Total Revenue

The total revenue (TR) of a seller equals the price of a good times the quantity of the good sold, P x Q = TR. For example, if the hamburger stand down the street sells 100 hamburgers today at $1.50 each, its total revenue is $150. Price elasticity of demand influences total revenue.

Demand is elastic: $P \uparrow \rightarrow TR \downarrow$

Demand is elastic: $P \downarrow \rightarrow TR \uparrow$



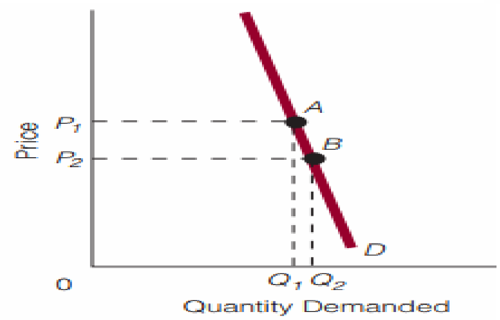
(a)

Between points A and B on the demand curve, demand is elastic. At point A, price is P_1 and quantity demanded is Q_1. Total revenue is equal to the rectangle $0P_1$ AQ_1 . Now, suppose we lower price to P_2. After the price decline, total revenue is now the rectangle $0P_2$ BQ_2 , which, as you can see, is larger than rectangle $0P_1$ AQ_1 . In other words, if demand is elastic and price declines, then total revenue will rise.

Of course, when price moves in the opposite direction, rising from P_2 to P_1, then the totalrevenue rectangle becomes smaller. In other words, if demand is elastic and price rises, then total revenue will fall.

Demand is inelastic : $P\uparrow \rightarrow TR\uparrow$

Demand is inelastic : $P\downarrow \rightarrow TR\downarrow$



(b)

In figure, demand is inelastic between points A and B on the demand curve. If we start at P_1 and lower price to P_2, then the total-revenue rectangle goes from $0P_1$ AQ_1 to the smaller rectangle $0P_2$ BQ_2 . In other words, if demand is inelastic and price falls, then total revenue will fall. Moving from the lower price, P_2, to the higher price, P_1, does just the opposite: If demand is inelastic and price rises, then the total revenue rectangle becomes larger; that is, total revenue rises.

Unit Elastic Demand and Total Revenue: If demand is unit elastic, the percentage change in quantity demanded equals the percentage change in price. That is, if price rises, then quantity demanded falls by the same percentage as the percentage rise in price. Total revenue does not change. If price falls, then quantity demanded rises by the same percentage as the percentage drop in price. Again, total revenue does not change. If demand is unit elastic, a rise or fall in price leaves total revenue unchanged.

Price Elasticity of Demand along a Straight-Line Demand Curve The price elasticity of demand for a straight-line downward-sloping demand curve varies from highly elastic to highly inelastic. At the midpoint it is unitary elastic, above the midpoint elastic and below the midpoint it is inelastic.

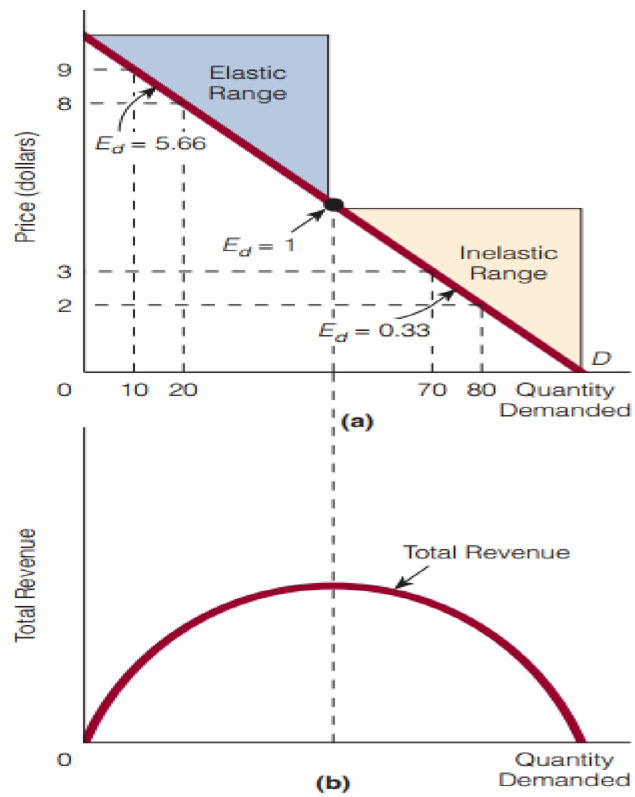


EXHIBIT 7

Consider the price elasticity of demand at the upper range of the demand curve in Exhibit 7(a). Whether the price falls from \$9 to \$8 or rises from \$8 to \$9, using the formula for price elasticity of demand, we calculate the price elasticity of demand as 5.66.

Suppose in the figure

P	Q_d

9	10
8	20

$$\Delta Q = 20 - 10 = 10$$
$$\Delta P = 8 - 9 = -1$$
$$P = \frac{9+8}{2} = 8.5$$
$$Q = \frac{10+20}{2} = 15$$

$$E_d = \frac{\Delta Q_d}{\Delta P}\frac{P}{Q_d} = \frac{20}{-1} x \frac{8.5}{15} = 5.66, elastic$$

Below the midpoint

P	Q_d
3	70
2	80

$$\Delta Q = 80 - 70 = 10$$
$$\Delta P = 2 - 3 = -1$$
$$P = \frac{3+2}{2} = 2.5$$
$$Q = \frac{70+80}{2} = 75$$

$$E_d = \frac{\Delta Q_d}{\Delta P}\frac{P}{Q_d} = \frac{10}{-1} x \frac{2.5}{75} = 0.33, inelastic$$

It is proved that elasticity of demand above the midpoint of a downward sloping demand curve is highly elastic and below the midpoint is highly inelastic. In other words, along the range of the demand curve identified, price elasticity goes from being greater than 1 (5.66) to being less than 1 (0.33).

The elastic and inelastic ranges along the straight line downward-sloping demand curve can be related to a total revenue curve [see Exhibit 7(b)]. If we start in the elastic range of the demand curve in Exhibit 7(a) and lower price, then total revenue rises, as shown in Exhibit 7(b). That is, as price is coming down within the elastic range of the demand curve in part (a), total revenue is rising in part (b).

When price has fallen enough that we move into the inelastic range of the demand curve in part (a), further price declines simply lower total revenue, as shown in part (b). Therefore, total revenue is at its highest—its peak—when price elasticity of demand equals 1.

Determinants of Price Elasticity of Demand

Four factors are relevant to the determination of price elasticity of demand:

1. Number of substitutes
2. Necessities versus luxuries
3. Percentage of one's budget spent on the good
4. Time

Because all four factors interact, we hold all other things constant as we discuss each factor.

1. **Number of Substitutes**

Suppose good A has 2 substitutes and good B has 15 substitutes.

• The more substitutes a good has, the higher the price elasticity of demand will be.

• The fewer substitutes a good has, the lower the price elasticity of demand will be.

• The more broadly defined the good, the fewer substitutes it will have.

• The more narrowly defined the good, the more substitutes it will have.

When the price of good A rises 10 percent, people can turn to 2 substitutes. The quantity demanded of good A falls, but not by as much as if 15 substitutes had been available, as there are for good B.

In other words, the quantity demanded will be more responsive to the 10 percent price rise for the good that has 15 substitutes good B than the good has less substitutes, good A.

2. **Necessities Versus Luxuries**

Generally, the more that a good is considered a luxury (a good that we can do without) rather than a necessity (a good that we cannot do without), the higher the price elasticity of demand will be. For example, consider two goods: jewelry and a medicine for controlling high blood pressure. If the price of jewelry rises, cutting back on purchases is easy. However, if the price of the medicine for controlling one's high blood pressure rises, cutting back is not so easy. We expect the price elasticity of demand for jewelry to be higher than that for high blood pressure medicine.

3. **Percentage of One's Budget Spent on the Good**

Claire Rossi has a monthly budget of $3,000. Of this monthly budget, she spends $3 per month on pens and $400 per month on dinners at restaurants. In terms of percentages, she spends 0.1 percent of her monthly budget on pens and 13 percent of her monthly budget on dinners at restaurants. Suppose both the price of pens and the price of dinners at restaurants double. Claire is likely to be more responsive to the change in the price of restaurant dinners than to the change in the price of pens. She feels the pinch of a doubling in the price of a good on which she spends 0.1 percent of her budget a lot less than a doubling in price of a good on which she spends 13 percent. So, Claire is more likely to ignore the increased price of pens than she is to ignore the heightened price of restaurant dinners.

• The greater the percentage of one's budget that goes to purchase a good, the higher the price elasticity of demand will be. • The smaller the percentage of one's budget that goes to purchase a good, the lower the price elasticity of demand will be.

4. Time

As time passes, buyers have greater opportunities to be responsive to a price change.

• The more time that passes (since the price change), the higher the price elasticity of demand for the good will be.

• The less time that passes, the lower the price elasticity of demand for the good will be.

In other words, the price elasticity of demand for a good is higher in the long run than in the short run.

OTHER ELASTICITY CONCEPTS

This section looks at three other elasticities:

- Cross elasticity of demand
- Income elasticity of demand
- Price elasticity of supply

Cross elasticity of demand measures the responsiveness in the quantity demanded of one good to changes in the price of another good. It is calculated by dividing the percentage change in the quantity demanded of one good by the percentage change in the price of another. That is,

$$E_c = \frac{\text{Percentage change in quantity demanded of one good}}{\text{Percentage change in price of another good}}$$

where E_c stands for the coefficient of cross elasticity of demand, or the elasticity coefficient.

Suppose that, when the price of Jiffincreases by 10 percent, the quantity demanded of Skippy increases by 45 percent. Then, the cross elasticity of demand for coffee with respect to the price of tea is

$$E_c = \frac{\text{Percentage change in quantity demanded of Skippy}}{\text{Percentage change in price of Jif}}$$

$$E_c = \frac{\Delta Q_s}{\Delta P_J} \frac{P_j}{Q_s}$$

$$E_c = \frac{45}{10} = 4.5$$

In this case, the cross elasticity of demand is a positive 4.5. When the cross elasticity of demand is positive, the percentage change in the quantity demanded of one good (the good mentioned in the numerator) moves in the same direction as the percentage change in the price of the other good (the good mentioned in the denominator). A positive cross elasticity of demand is a characteristic of goods that are substitutes. As the price of Jif rises, the demand curve for Skippy shifts rightward, causing the quantity demanded of Skippy to increase at every price. So, if $E_c > 0$, the two goods are substitutes:

$E_c > 0 \rightarrow$ Goods are substitutes

If the elasticity coefficient is negative ($E_c < 0$), then the two goods are complements:

$E_c > 0 \rightarrow$ Goods are complements

A negative cross elasticity of demand occurs when the percentage change in the quantity demanded of one good (the good mentioned in the numerator) and the percentage change in the price of another good (the good mentioned in the denominator) move in opposite directions. For example, suppose the price of cars increases by 5 percent and the quantity demanded of car tires decreases by 10 percent. Calculating the cross elasticity of demand, we have -10 percent ÷ 5 percent = -2. Thus, cars and car tires are complements.

Income Elasticity of Demand

Income elasticity of demand measures the responsiveness of quantity demanded to changes in income. It is calculated by dividing the percentage change in quantity demanded of a good by the percentage change in income. That is,

$$E_y = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

$$= \frac{\Delta Q}{\Delta Y}\frac{Y}{Q}$$

where E_y denotes the coefficient of income elasticity of demand, or the elasticity coefficient.

Income elasticity of demand is positive (0 Ey .) for a normal good. Recall that a normal good is a good whose demand, and thus whose quantity demanded, increases, given an increase in income. Hence, for a normal good, the variables in the numerator and denominator in the formula for the income elasticity of demand move in the same direction:

$$E_y > 0 \rightarrow \text{Normal good}$$

In contrast to a normal good, the demand for an inferior good decreases as income increases.

Income elasticity of demand for an inferior good is negative ($E_y > 0$):

$$E_y < 0 \rightarrow \text{Inferior good}$$

Now, suppose a person's income increases from \$500 to \$600 per month and, as a result, the quantity demanded of good X increases from 20 units to 30 units per month. Then we have income elasticity of 2.2, as shown here:

$$E_y = \frac{\frac{10}{25}}{\frac{100}{550}} = 2.2$$

Since E_y is a positive number, so good X is a normal good. Also,

- Because $E_y > 1$, demand for good X is said to be **income elastic**. In other words, the percentage change in quantity demanded of the good is greater than the percentage change in the purchaser's income.

- If $E_y < 1$, the demand for the good is said to be **income inelastic**.
- If $E_y = 1$, then the demand for the good is **income unit elastic**.

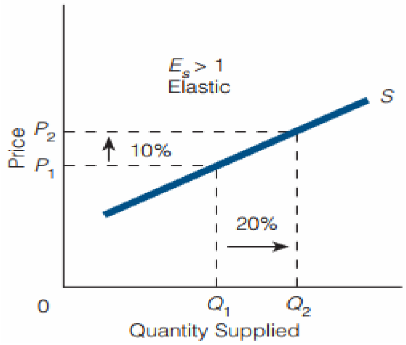
Price Elasticity of Supply Price elasticity of supply measures the responsiveness of quantity supplied to changes in price. It is calculated by dividing the percentage change in the quantity supplied of a good by the percentage change in the price of the good. Mathematically,

$$E_s = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

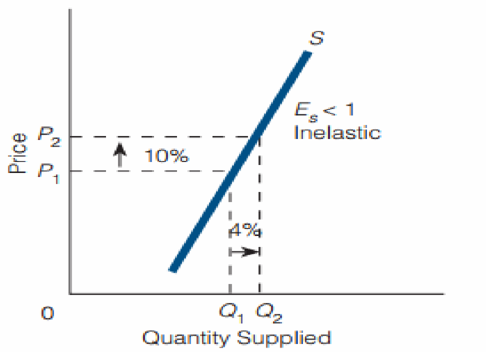
Where, E_s stands for the coefficient of price elasticity of supply, or the elasticity coefficient.

Supply can be classified as elastic, inelastic, unit elastic, perfectly elastic, or perfectly inelastic:

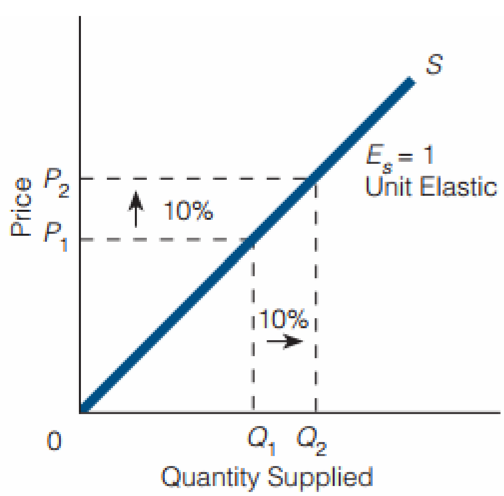
• Elastic supply ($E_s > 1$: a percentage change in quantity supplied that is greater than the percentage change in price of the good.



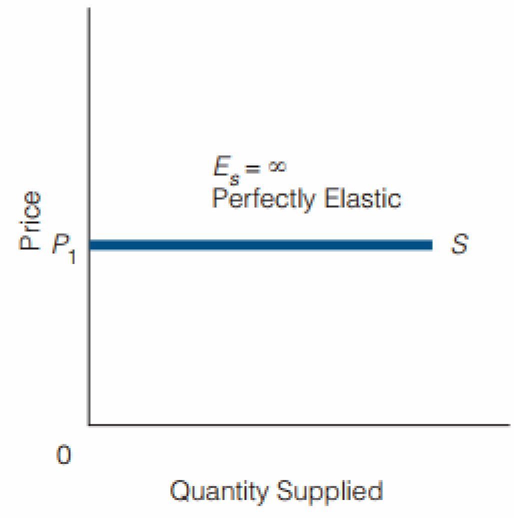
• Inelastic supply (E_s >1): a percentage change in quantity supplied that is less than the percentage change in price of the good.



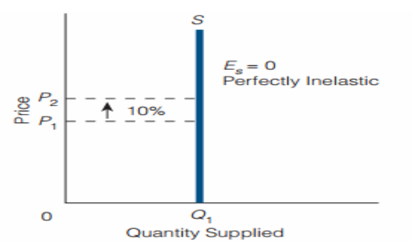
• Unit elastic supply (E_s = 1) : a percentage change in quantity supplied that is equal to the percentage change in price of the good.



• Perfectly elastic supply ($E_s = \infty$) : a small change in price changes the quantity supplied by an infinitely large amount (and thus the supply curve, or a portion of the overall supply curve, is horizontal).



•Perfectly inelastic supply (E_s= 0): a change in price brings no change in quantity supplied (and thus the supply curve, or a portion of the overall supply curve, is vertical).



Price Elasticity of Supply and Time

For goods whose quantity supplied can increase with time—a characteristic of most goods (but not, e.g., original Picasso paintings)—the longer the period of adjustment is to a change in price, the higher the price elasticity of supply will be. The obvious reason is that additional production takes time or may be impossible.

EXHIBIT 9

Summary of the Four Elasticity Concepts

Type	Calculation	Possibilities	Terminology
Price elasticity of demand	$\frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$	$E_d > 1$	Elastic
		$E_d < 1$	Inelastic
		$E_d = 1$	Unit elastic
		$E_d = \infty$	Perfectly elastic
		$E_d = 0$	Perfectly inelastic
Cross elasticity of demand	$\frac{\text{Percentage change in quantity demanded of one good}}{\text{Percentage change in price of another good}}$	$E_c < 0$	Complements
		$E_c > 0$	Substitutes
Income elasticity of demand	$\frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$	$E_y > 0$	Normal good
		$E_y < 0$	Inferior good
		$E_y > 1$	Income elastic
		$E_y < 1$	Income inelastic
		$E_y = 1$	Income unit elastic
Price elasticity of supply	$\frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$	$E_s > 1$	Elastic
		$E_s < 1$	Inelastic
		$E_s = 1$	Unit elastic
		$E_s = \infty$	Perfectly elastic
		$E_s = 0$	Perfectly inelastic

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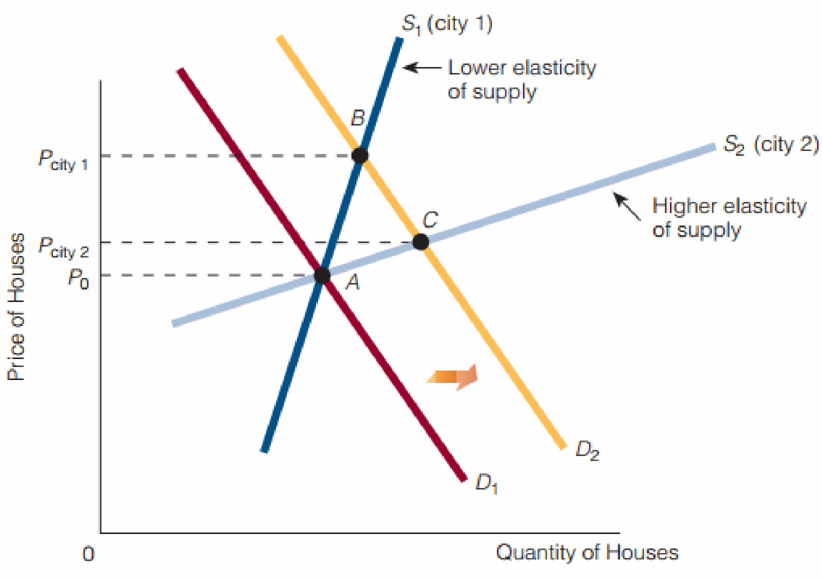
House Prices and Elasticity of Supply

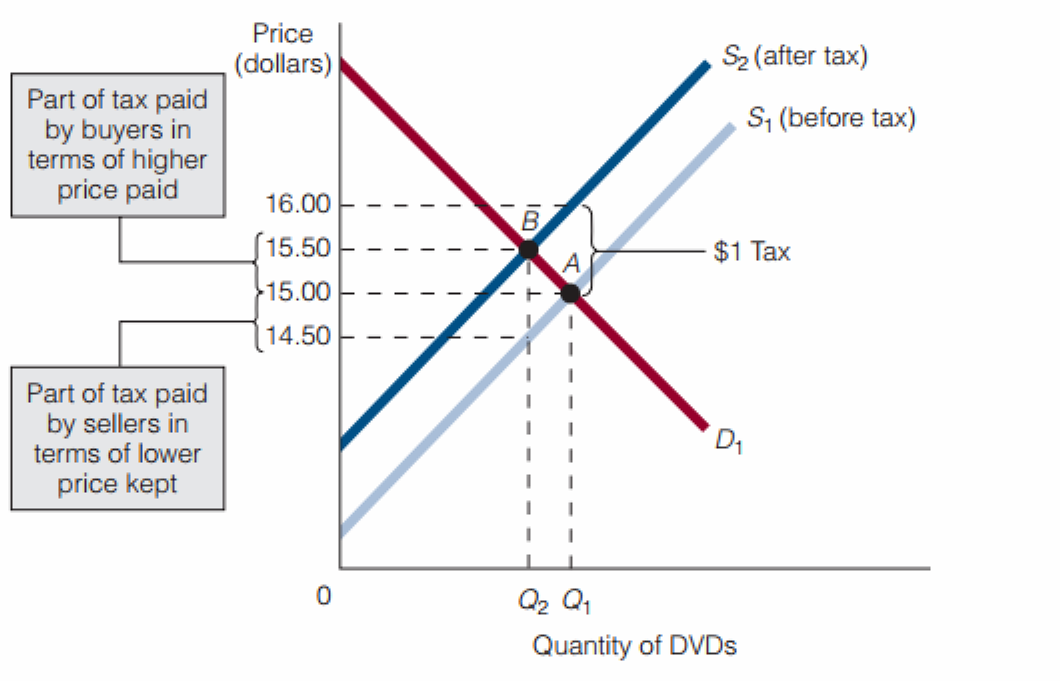


S_1 represents the supply of housing in city 1, and S_2 represents the supply of housing in city 2. S_1 has lower elasticity of supply than S_2. Suppose the demand for housing in each city rises from D_1 to D_2. As a result, the price of houses rises in both cities but it rises by more in city 1 than city 2. In other words, the lower the elasticity of supply, the greater is the increase in price.

Who Pays the Tax? Many people think that, if government places a tax on the seller of a good, the seller actually pays the tax. However, the placement of a tax is not the same as its payment, and placement does not guarantee payment.

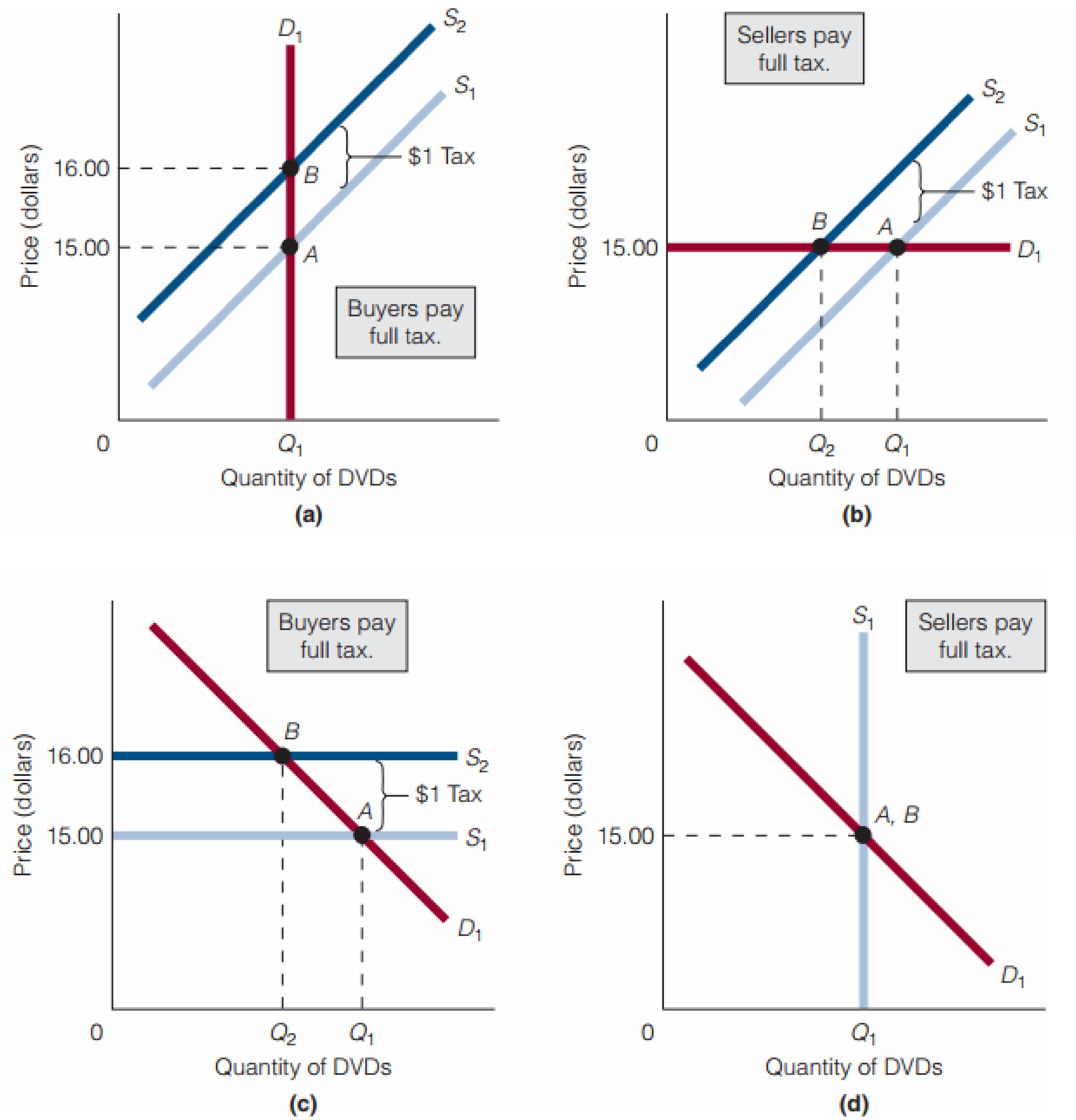
Before the tax is imposed, the equilibrium price and the quantity of tapes are $15 and Q_1, respectively. The tax per DVD shifts the supply curve leftward from S_1 to S_2 and raises the equilibrium price from $15.00 to $15.50. The vertical distance between the two supply curves represents the $1-per-DVD tax, because what matters to sellers is how much they keep for each DVD sold, not how much buyers pay. Part of the tax is paid by buyers through a higher price paid ($15.50 instead of $15.00), and part of the tax is paid by sellers through a lower price kept ($14.50 instead of $15.00).

Before the tax, buyers pay $15.00. After the tax, buyers pay $15.50.Before the tax, sellers receive $15.00 and keep $15.00. After the tax, sellers receive $15.50 and keep $14.50. So, although the full tax was placed on the sellers, they paid only one-half of it; although none of the tax was placed on buyers, they paid one-half of it too. The lesson is that government can place a tax on whomever it wants, but the laws of supply and demand determine who actually ends up paying it.



Elasticity and the Tax

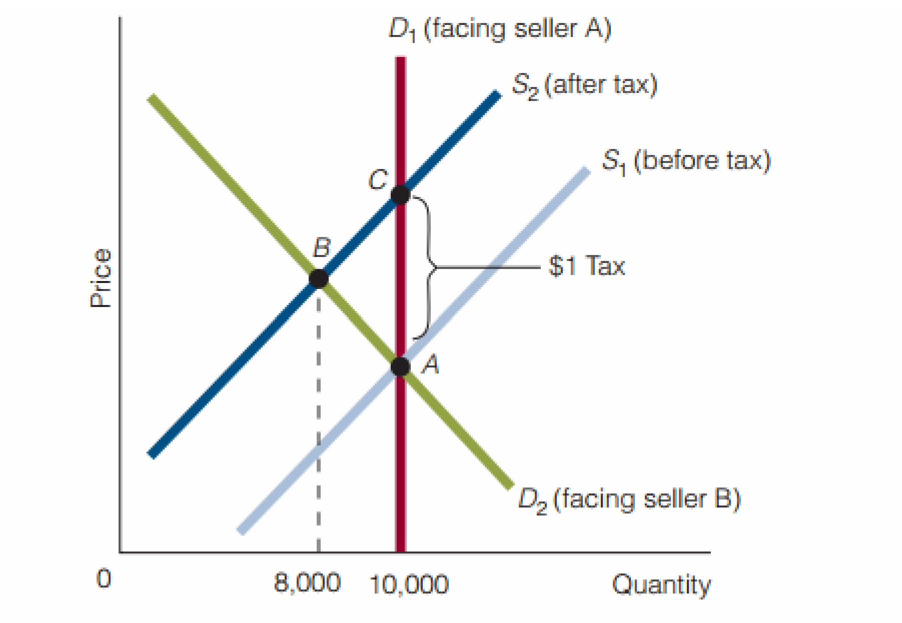
In our tax example, buyers paid half of the $1 tax and sellers paid half, but that is not the outcome in every situation. In some situations, the buyer can pay more than half the tax. In fact, the buyer can pay the full tax if demand for the good is perfectly inelastic, figure(a). Here, the tax shifts the supply curve from S_1 to S_2, and the equilibrium price rises from $15.00 to $16.00. In other words, if demand is perfectly inelastic and a tax is placed on the sellers of a good, buyers pay the full tax as part of a higher price. Parts (b)–(d) show other cases. In part (b), demand is perfectly elastic. The tax shifts the supply curve from S_1 to S_2, but the equilibrium price does not change. Sellers must therefore pay the full tax if demand is perfectly elastic. In part (c), supply is perfectly elastic, and buyers pay the full tax. In part (d), a change in price causes no change in quantity supplied. If sellers try to charge a higher price than $15 for their good (and thus try to get buyers to pay some of the tax), a surplus will result, driving the price back down to $15. In this case, sellers pay the full tax. Although the exhibit does not show it, sellers would receive $15, turn over $1 to the government, and keep $14 for each unit sold.



Degree of Elasticity and Tax Revenue

Suppose that, of two sellers, seller A faces a perfectly inelastic demand for her product and is currently selling 10,000 units a month. Suppose also that seller B faces an elastic demand for his product and is currently selling 10,000 units a month. Government is thinking about placing a $1 tax per unit of product sold on one of the two sellers. If government's objective is to maximize tax revenues, it should tax seller A, because that seller is facing the inelastic demand curve.

Exhibit 13



In Exhibit 13, the demand curve facing seller A is D_1 and the demand curve facing seller B is D_2. S_1 represents the supply curve for both firms. Currently, both firms are at equilibrium at point A, selling 10,000 units. If government places a $1 tax per unit sold on seller A, then the supply curve shifts to S_2 and the equilibrium is now at point C. Because demand is perfectly inelastic, seller A still sells 10,000 units and the government's tax revenue from seller A equals the tax ($1) times 10,000 units, or $10,000. If government places the $1 tax per unit sold on seller B, then tax revenue will be only $8,000 Because, when the tax shifts the supply curve to S_2, the equilibrium moves to point B, at which only 8,000 units are sold.

The lesson is that, given the $1 tax per unit sold, tax revenues are maximized by placing the tax on the seller who faces the more inelastic (less elastic) demand curve.

Sample questions for exam. From this chapter

1. **Derive formula for the elasticity of Demand.**
2. **Determine the elasticity of demand using following information. Comment on your result and draw a demand curve for this estimated elasticity of demand.**

Suppose,

P	Q_d
2	10
1	25

3. **Draw demand curve for different types of elasticity of demand and interpret.**
4. **Describe the relationship between price elasticity of demand and total revenue.**
5. **Prove that elasticities of demand at different points on a straight line demand curve are different.**
6. **Explain the relationship between elasticity and the tax revenue**